

IPFS: A Decentralised Cloud and File System for the Blockchain Environment

InterPlanetary File System (IPFS) is a dedicated protocol for storing and sharing data in decentralised applications like blockchain. It seeks to make the Internet more free, open, secure, and high performance. It works on the Go and JavaScript programming languages. Discover more about IPFS in this article.

The smart gadgets in use today run on high performance operating systems like Android, iOS, Windows, Blackberry, Tizen, Lune OS, Kai OS, and many others. With the increased use of such platforms, there is a greater need for security and integrity, for which blockchain technologies are now being adopted.

In classical Web applications, a centralised Web server is deployed. Centralised Web apps with a single server, cloud or data centre are deployed for many services including e-governance, medical records management, academic institutional management, the corporate sector, etc. There are numerous vulnerabilities in these traditional Web based applications deployed on centralised Web servers.

In these scenarios, all users or remote systems access the Web services using their Web client application, which includes assorted Web browsers. The major problem in this type of environment is the centralisation; so if the server fails or is hacked, the entire network environment gets compromised.

In deployments in which only one server or a few more replicated servers are hacked, all data can be leaked out, copied or destroyed, and this is the major issue with centralised servers. To avoid this, there is a

specific implementation that uses a decentralised Web.

In a decentralised Web, there is no central repository of data or applications. Blockchain technology only uses a decentralised Web. The data and applications are stored at multiple places and regions using blockchain technology to ensure security and privacy.

Blockchain technology deploys decentralised apps or dApps

In a decentralised Web, the data is stored on every node or system using that network. In the blockchain environment, there is no threat of data loss because

the chain of nodes (blockchain) is immutable (unchangeable) and all blocks are interconnected with a dynamic hash (encrypted text). The blocks in the blockchain environment are programmed with high performance protocols so that they become immutable and cannot be hacked or modified by any third party intruders or sniffers.

Decentralised platforms for blockchain implementations

There are hundreds of distributed cloud platforms that can be used for blockchains. Table 1 features the main decentralised platforms for blockchain

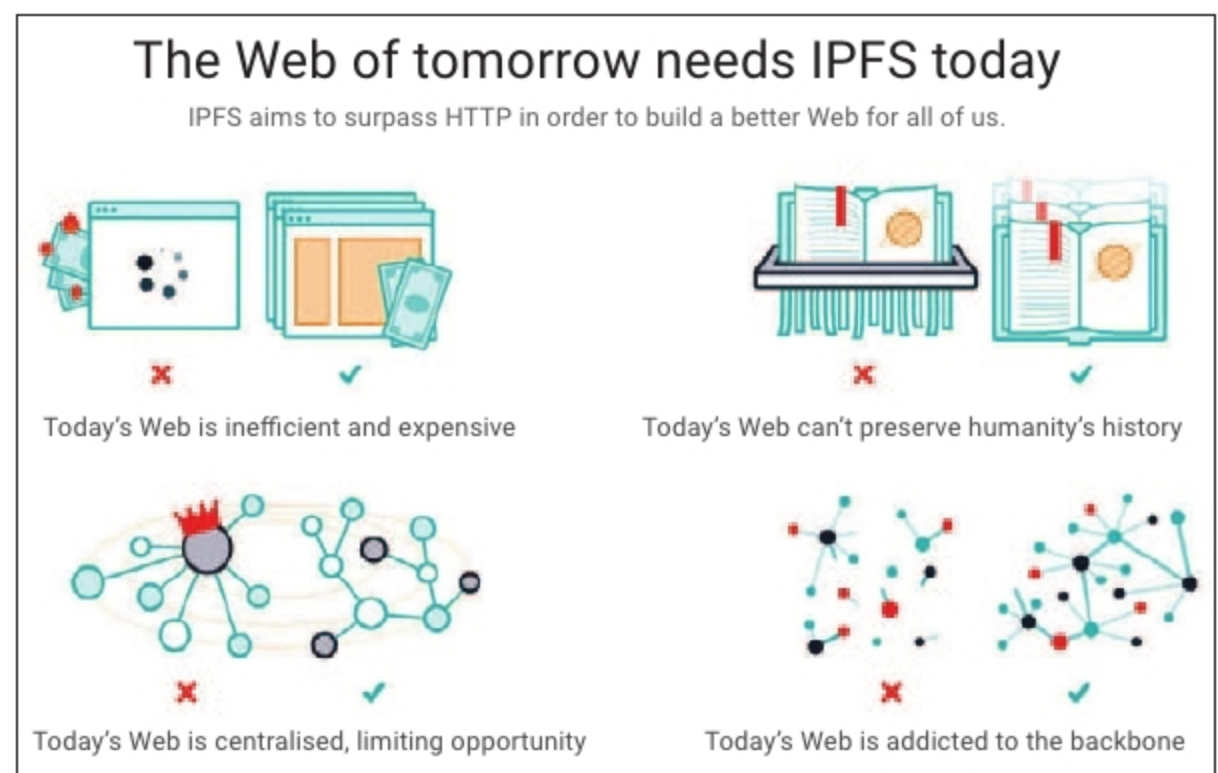


Figure 1: Advantages of IPFS over HTTP (Source: Official portal of IPFS - <https://ipfs.io/>)

